

Do or Die

Analyzing the NFL's Two-Minute Drill

Parker Merritt



Abstract

NFL Games often come down to the final two minutes. In the case that a team gets the ball late, the offense usually changes play calling and strategy due to time constraints on the drive. We analyze timeout value in two-minute scenarios using the NFL's Play-By-Play Data in the programming language R.



Definition

Two Minute Drill - A scenario where an offensive team's first play from scrimmage takes place between 0:13-2:30 game time left, and the offensive team is tied or down by eight or fewer points.



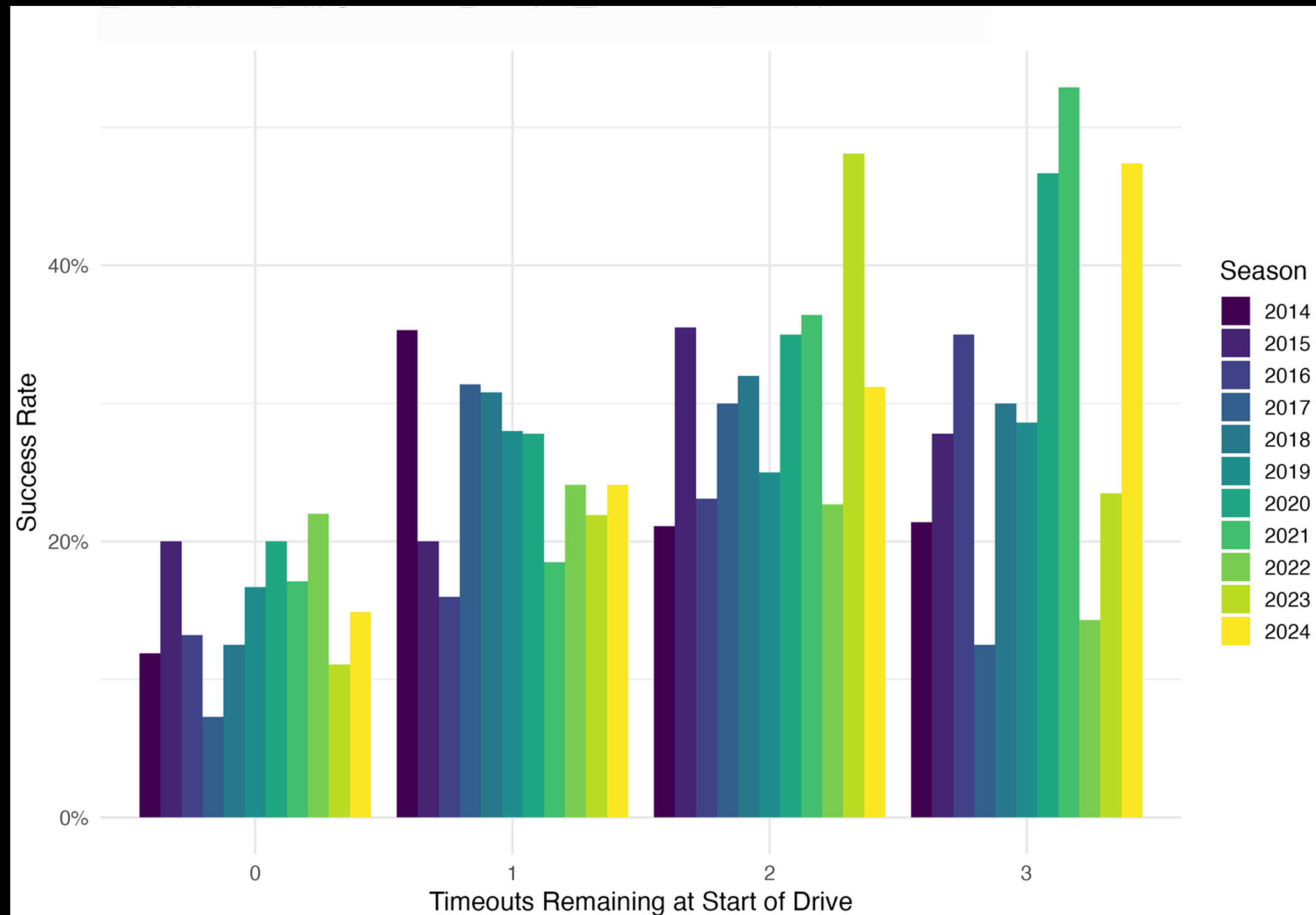
Timeouts

A team receives 3 timeouts per half. A timeout pauses play and creates space for a coach to talk with their team, give their players rest, and/or simply to stop the clock. We are especially interested in this last feature, for after a player is tackled within the boundary markers, the clock doesn't stop. This means a coach can pause the clock even if a player is tackled within the boundary.



Historical Trends

2-Min Drill Success Rate by Timeouts Remaining (2014-2024)



The success rate relative to timeouts remaining is inconsistent over time. However, this is likely due to the variance between the average time remaining for the 2-minute drill across timeout values.



Historical Trends

2-Min Drill vs. Timeouts (2000-2024)

Timeouts Remaining	Oppurtunities	Avg. Success Rate	Avg. Start Time (seconds remaining)
0	1084	0.152	72.5
1	619	0.262	89.3
2	567	0.279	86.7
3	355	0.293	92.2

This is confirmed in our analysis across all years, as there is much variance in the average start time relative to timeouts remaining.



Analysis

Log Regression Model 0-3 timeouts

Term	Estimate	Std. Error	p-Value	Odds Ratio	% Change in Odds
Timeouts Remaining	0.239	0.044	4.9×10^{-8}	1.27	27.00%
Start Time	0.356	0.05	1.1×10^{-12}	1.43	42.70%



Log Regression Model 0-1 timeouts

Term	Estimate	Std. Error	p-Value	Odds Ratio	% Change in Odds
Timeouts Remaining	0.53	0.128	3.5×10^{-5}	1.7	69.90%
Start Time	0.394	0.0662	2.7×10^{-9}	1.48	48.20%

Log Regression Model 1-3 timeouts

Term	Estimate	Std. Error	p-Value	Odds Ratio	% Change in Odds
Timeouts Remaining	0.0716	0.0736	0.331	1.07	7.40%
Start Time	0.272	0.0595	4.7×10^{-6}	1.31	31.30%

Analysis

There are statistically significant results for 0-3 timeouts remaining, but this significance disappears once we partition on one timeout. This suggests that having one timeout is key for a two-minute drill, but having multiple timeouts might not be as useful as once thought. This could be because with one timeout, defenses still must respect the run. Alternative explanations could be that our dataset is too large (2000-2024), and that we get statistically insignificant results for single years due to the relatively small sample size and the time remaining between opportunities.

Also note that the individual years had no statistically significant impact on the success rate (ex. 2024 vs. 2021).



Future Work

We would like to analyze whether preventing a false start via a timeout in the third quarter is worth avoiding a penalty, or is it more valuable to lose five yards but still have a timeout.

Doing this analysis per team would be interesting as well, for some coaches have different styles in 2-minute drills offensively and defensively.



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parkermeritt05@gmail.com

www.linkedin.com/in/parkermeritt05

